

Fraction $\Rightarrow$ A part of a whole
 { पूर्ण का एक हिस्सा }

like, $\frac{2}{6} \rightarrow$ part
 $6 \rightarrow$ whole

$x \rightarrow$ upper part is called Numerator
 $y \rightarrow$ lower part is called Denominator

Q1. $\frac{132}{198} = ?$

- ☒ A) $\frac{2}{3}$ B) $\frac{3}{2}$ C) $\frac{4}{5}$ D) $\frac{11}{15}$ E) $\frac{5}{6}$

$\frac{132}{198} \rightarrow$ Numerator
 $198 \rightarrow$ Denominator

$$= \frac{132}{198} = \frac{11 \times 4 \times 3}{11 \times 6 \times 3} = \frac{4}{6} = \frac{2}{3}$$

Q2. $-\frac{150}{210} = ?$

- A) $-\frac{7}{5}$ B) $\frac{5}{7}$ ☒ C) $-\frac{5}{7}$ D) $-\frac{3}{7}$ E) $-\frac{2}{3}$

$$-\frac{150}{210} = -\left(\frac{5 \times 3}{7 \times 3}\right) = -\frac{5}{7}$$

Q3. $\frac{5}{12} = \frac{?}{96}$

- A) 32 B) 36 C) 44 ☒ D) 40 E) 30

$$\frac{5}{12} = \frac{x}{96}$$

$$\frac{5 \times 96}{12} = x$$

$$x = 40$$

Q4. $\frac{?}{45} = \frac{7}{15}$

- A) 14 B) 18 C) 27 D) 24 ☒ E) 21

$$\frac{x}{45} = \frac{7}{15}$$

$$x = \frac{7}{15} \times 45$$

$$x = 21$$

Q5. $\frac{9}{?} = \frac{3}{14}$

A) 28

~~B) 42~~

C) 36

D) 48

E) 54

$$\frac{9}{x} = \frac{3}{14}$$

$$x = \frac{9}{3} \times 14$$

$$x = 3 \times 14$$

$$x = 42$$

Q6. If $x \times \left(\frac{8}{9}\right) = \frac{16}{27}$, then $x = ?$

A) $1/3$ ~~B) $2/3$~~ C) $3/2$ D) $9/8$ E) $8/9$

$$x \times \frac{8}{9} = \frac{16}{27}$$

$$x = \frac{16}{27} \times \frac{9}{8} = \frac{16}{8} \times \frac{9}{27} = 2 \times \frac{1}{3} = \frac{2}{3}$$

Q7. If $\frac{a}{b} = \frac{4}{9}$ and $a = 28$, then $b = ?$

A) 56

B) 84

C) 72

~~D) 63~~

E) 81

$$\frac{a}{b} = \frac{4}{9}, \quad \frac{28}{b} = \frac{4}{9}$$

$$\{a = 28\}$$

$$b = \frac{28}{4} \times 9$$

$$b = 7 \times 9$$

$$b = 63$$

Mixed $\leftrightarrow$ Improper

Q8. $4\frac{7}{9} = ?$

A) $41/9$ B) $42/9$ ~~C) $43/9$~~ D) $44/9$ E) $45/9$

$$4\frac{7}{9} = 4 + \frac{7}{9} = \frac{36+7}{9} = \frac{43}{9}$$

OR

$$4\frac{7}{9} = \frac{(4 \times 9) + 7}{9} = \frac{36+7}{9} = \frac{43}{9}$$

Numerator
Denominator.

when

Numerator < Denominator

It's Proper fraction.

like, $\frac{2}{3}, \frac{3}{7}, \frac{9}{10}, \frac{11}{25}$ etc.

when

Numerator > Denominator.

It's Improper fraction

like, $\frac{37}{7}, \frac{11}{5}, \frac{61}{2}$ etc.

Mixed Fraction $\Rightarrow$ Fraction + Integer.

Q9. $6\frac{5}{12} = ?$

{ Improper Fraction > 1 }

A) $\frac{77}{12}$

B) $\frac{75}{12}$

C) $\frac{79}{12}$

D) $\frac{72}{12}$

E) $\frac{71}{12}$

$$6\frac{5}{12} = 6 + \frac{5}{12}$$

$$\frac{(12 \times 6) + 5}{12} = \frac{72 + 5}{12}$$

$$= \frac{77}{12}$$

OR

$$6\frac{5}{12} = \frac{(6 \times 12) + 5}{12} = \frac{72 + 5}{12} = \frac{77}{12}$$

Q10. $\frac{58}{7} = ?$

A) $7\frac{2}{7}$

B) $8\frac{1}{7}$

C) $8\frac{2}{7}$

D) $8\frac{3}{7}$

E) $9\frac{2}{7}$

$\frac{58}{7} \rightarrow$ Improper Fraction

$= 8\frac{2}{7}$ { Mixed Fraction }

8 $\rightarrow$ Quotient
Divisor $\leftarrow 7$ $\sqrt{58}$
56
2 $\rightarrow$ Remainder

Q11. $-\frac{73}{8} = ?$

A) $-10\frac{1}{8}$

B) $-8\frac{1}{8}$

C) $-9\frac{5}{8}$

D) $9\frac{1}{8}$

E) $-9\frac{1}{8}$

$-\frac{73}{8}$ { Improper Fraction }

$= -(9\frac{1}{8})$ Mixed Fraction

9
8 $\sqrt{73}$
72
1

Q12. Find the greater fraction

$$\left(\frac{7}{12}, \frac{5}{9}\right)$$

$$\frac{7}{12}, \frac{5}{9}$$

Method 1:- Use Cross Multiplication

$$\frac{7}{12} \times \frac{5}{9}$$

$$63 > 60$$

$$\text{So, } \frac{7}{12} > \frac{5}{9}$$

OR

Method 2:-

$$\frac{7}{12}, \frac{5}{9}$$

Denominators को equal करना है!

$$\frac{7 \times 3}{12 \times 3}, \frac{5 \times 4}{9 \times 4}$$

$$\frac{21}{36} > \frac{20}{36}$$

$$\text{So, } \frac{7}{12} > \frac{5}{9}$$

Q13. Find the smaller fraction

$$\left(\frac{11}{15}, \frac{3}{4}\right)$$

$$\frac{11}{15} \times \frac{3}{4}$$

$$(11 \times 4) \quad (3 \times 15)$$

$$44 < 45$$

OR

$$\frac{11 \times 4}{15 \times 4}, \frac{3 \times 15}{4 \times 15}$$

$$\frac{44}{60} < \frac{45}{60}$$

Q. Find the larger fraction

$$\left(\frac{22}{7}, \frac{28}{9}\right)$$

$$\frac{22}{7} \times \frac{28}{9}$$

$$(22 \times 9) \quad (28 \times 7)$$

$$198 > 196$$

OR

$$\frac{22 \times 9}{7 \times 9}, \frac{28 \times 7}{9 \times 7}$$

$$\frac{198}{63} > \frac{196}{63}$$

$$\text{So, } \frac{22}{7} > \frac{28}{9}$$

Let's understand Method 2 in detail :-

$$\frac{7}{12}, \frac{5}{9}$$

We need to have same denominators on both sides.

$$\left(\frac{7 \times 3}{12 \times 3}\right), \left(\frac{5 \times 4}{9 \times 4}\right) \rightarrow$$

Den. & Num.
same Number
में Multiply konge

$$\frac{21}{36} > \frac{20}{36}$$

Then, Numerator ke basis par, smaller & larger decide hoga

Q14. Put in ascending order

$$\frac{2}{5}, \frac{3}{7}, \frac{5}{12}$$

A) $\frac{2}{5} < \frac{3}{7} < \frac{5}{12}$

B) $\frac{3}{7} < \frac{5}{12} < \frac{2}{5}$

C) $\frac{5}{12} < \frac{2}{5} < \frac{3}{7}$

☒ D) $\frac{2}{5} < \frac{5}{12} < \frac{3}{7}$

E) $\frac{5}{12} < \frac{3}{7} < \frac{2}{5}$

Answer = $\frac{2}{5} < \frac{5}{12} < \frac{3}{7}$

$$\frac{2}{5} \quad \frac{3}{7} \quad \frac{5}{12}$$

$$\frac{2}{5} \times \frac{3}{7} \quad \left\{ \frac{3}{7} > \frac{2}{5} \right\}$$

$$14 < 15$$

$$\frac{3}{7} \times \frac{5}{12} \quad \left\{ \frac{3}{7} > \frac{5}{12} \right\}$$

$$36 > 35$$

$$\frac{2}{5} \times \frac{5}{12} \quad \left\{ \frac{5}{12} > \frac{2}{5} \right\}$$

$$25 > 24$$

Hence, $\frac{3}{7} > \frac{5}{12} > \frac{2}{5}$

Q15. Put in descending order

$$\frac{7}{8}, \frac{11}{12}, \frac{9}{10}$$

A) $\frac{9}{10} > \frac{11}{12} > \frac{7}{8}$

☒ B) $\frac{11}{12} > \frac{9}{10} > \frac{7}{8}$

C) $\frac{7}{8} > \frac{9}{10} > \frac{11}{12}$

D) $\frac{11}{12} > \frac{7}{8} > \frac{9}{10}$

E) $\frac{7}{8} > \frac{11}{12} > \frac{9}{10}$

Answer = $\frac{11}{12} > \frac{9}{10} > \frac{7}{8}$

$$\frac{7}{8} \quad \frac{11}{12} \quad \frac{9}{10}$$

$$\frac{7}{8} \times \frac{11}{12} \quad \left\{ \frac{7}{8} < \frac{11}{12} \right\}$$

$$84 < 88$$

$$\frac{11}{12} \times \frac{9}{10} \quad \left\{ \frac{11}{12} > \frac{9}{10} \right\}$$

$$110 > 108$$

$$\frac{7}{8} \times \frac{9}{10} \quad \left\{ \frac{7}{8} < \frac{9}{10} \right\}$$

$$70 < 72$$

Hence, $\frac{11}{12} > \frac{9}{10} > \frac{7}{8}$

LCM → Least Common Multiple

ऐसा Lowest (Minimum) number जो given numbers के Table पर common है!

Q16. $\frac{7}{18} + \frac{5}{18} = ?$

A) $\frac{2}{3}$

B) $\frac{11}{18}$

C) $\frac{5}{6}$

D) $\frac{5}{9}$

E) $\frac{13}{18}$

$$\frac{7}{18} + \frac{5}{18} = \frac{7+5}{18}$$

$$= \frac{12}{18} = \frac{6 \times 2}{6 \times 3} = \frac{2}{3}$$

Q17. $\frac{13}{25} - \frac{7}{25} = ?$

A) $\frac{8}{25}$

B) $\frac{5}{25}$

C) $\frac{6}{25}$

D) $\frac{12}{25}$

E) $\frac{1}{5}$

$$\frac{13}{25} - \frac{7}{25} = \frac{13-7}{25}$$

$$= \frac{6}{25}$$

Q18. $\left(\frac{11}{12} - \frac{5}{12}\right) - \frac{3}{12} = ?$

A) $\frac{1}{12}$

B) $\frac{1}{6}$

C) $\frac{1}{3}$

D) $\frac{1}{4}$

E) $\frac{5}{12}$

$$\left(\frac{11-5}{12}\right) - \frac{3}{12}$$

$$= \frac{6}{12} - \frac{3}{12} = \frac{6-3}{12} = \frac{3}{12} = \frac{1}{4}$$

Q19. $\left(-\frac{7}{20}\right) + \frac{11}{20} = ?$

- A) $\frac{1}{2}$ B) $\frac{2}{5}$ C) $\frac{3}{10}$ ☒ D) $\frac{1}{5}$ E) $\frac{1}{4}$

$$\begin{aligned}\left(-\frac{7}{20}\right) + \frac{11}{20} &= \frac{-7 + 11}{20} \\ &= \frac{+4}{20} = \frac{1}{5}\end{aligned}$$

Q20. $3\frac{4}{9} + 1\frac{2}{9} = ?$

- A) $4\frac{5}{9}$ ☒ B) $4\frac{2}{3}$ C) $5\frac{6}{9}$ D) $5\frac{2}{9}$ E) $4\frac{8}{9}$

$$\begin{aligned}\left(3\frac{4}{9}\right) + \left(1\frac{2}{9}\right) &= \left(3 + \frac{4}{9}\right) + \left(1 + \frac{2}{9}\right) = 3 + \frac{4}{9} + 1 + \frac{2}{9} \\ &= (3 + 1) + \left(\frac{4 + 2}{9}\right) = 4 + \frac{6}{9} = 4 + \frac{2}{3} = 4\frac{2}{3}\end{aligned}$$

Q21. $\frac{7}{18} + \frac{5}{12} = ?$

- ☒ A) $\frac{29}{36}$ B) $\frac{23}{36}$ C) $\frac{31}{36}$ D) $\frac{17}{24}$ E) $\frac{11}{18}$

LCM of 18 and 12

$$18 \times 2 = 36$$

$$12 \times 3 = 36$$

$$\frac{7}{18} + \frac{5}{12}$$

$$\frac{(7 \times 2) + (5 \times 3)}{36} = \frac{14 + 15}{36} = \frac{29}{36}$$

Q22. $\frac{13}{20} - \frac{7}{25} = ?$

- A) $\frac{49}{100}$ B) $\frac{53}{100}$ C) $\frac{37}{100}$ D) $\frac{17}{50}$ E) $\frac{7}{20}$

LCM of 20 and 25
 $20 \times 5 = 100$
 $25 \times 4 = 100$

$$\frac{13}{20} - \frac{7}{25} = \frac{(13 \times 5) - (7 \times 4)}{100} = \frac{65 - 28}{100} = \frac{37}{100}$$

Q23. $\left(\frac{5}{6} - \frac{1}{4}\right) - \frac{1}{3} = ?$

- A) $\frac{1}{6}$ B) $\frac{5}{12}$ C) $\frac{1}{3}$ D) $\frac{1}{4}$ E) $\frac{7}{12}$

Answer = $\frac{1}{4}$

LCM of 6, 4 and 3
 $6 \times 2 = 12$
 $4 \times 3 = 12$
 $3 \times 4 = 12$

$$\begin{aligned} &= \frac{5}{6} - \frac{1}{4} - \frac{1}{3} \\ &= \frac{(5 \times 2) - (1 \times 3) - (1 \times 4)}{12} \\ &= \frac{10 - 3 - 4}{12} = \frac{3}{12} = \frac{1}{4} \end{aligned}$$

Q24. $2\frac{1}{3} + 1\frac{5}{6} = ?$

- A) $3\frac{1}{6}$ B) $4\frac{1}{6}$ C) $4\frac{5}{6}$ D) $3\frac{5}{6}$ E) $5\frac{1}{6}$

$$\begin{aligned} &= 2\frac{1}{3} + 1\frac{5}{6} \\ &= \left(2 + \frac{1}{3}\right) + \left(1 + \frac{5}{6}\right) = 2 + \frac{1}{3} + 1 + \frac{5}{6} \end{aligned}$$

$$= (2 + 1) + \left(\frac{(1 \times 2) + (5 \times 1)}{6}\right)$$

{ LCM of 3 and 6 will be 6 }

$$= 3 + \left(\frac{2 + 5}{6}\right)$$

$$= 3 + \frac{7}{6} = 3 + \left(1 + \frac{1}{6}\right) = 4\frac{1}{6}$$

Q25. $5 - \frac{17}{12} = ?$

- A) $3\frac{5}{12}$ B) $3\frac{11}{12}$ C) $4\frac{7}{12}$ D) $2\frac{7}{12}$ ☒ E) $3\frac{7}{12}$

$$= \frac{5}{1} - \frac{17}{12}$$

$$= \frac{(5 \times 12) - (17 \times 1)}{12} = \frac{60 - 17}{12} = \frac{43}{12} = 3\frac{7}{12}$$

Q26. $(2\frac{3}{4}) - (1\frac{5}{8}) = ?$

- ☒ A) $1\frac{1}{8}$ B) $1\frac{3}{8}$ C) $1\frac{5}{8}$ D) $\frac{9}{8}$ E) $\frac{7}{8}$

$$= (2 + \frac{3}{4}) - (1 + \frac{5}{8}) = 2 + \frac{3}{4} - 1 - \frac{5}{8}$$

$$= (2 - 1) + (\frac{3}{4} - \frac{5}{8}) = 1 + \frac{(3 \times 2) - (5 \times 1)}{8} = 1 + \frac{1}{8} = 1\frac{1}{8}$$

Q27. $(\frac{21}{32}) \times (\frac{16}{63}) = ?$

- A) $\frac{2}{7}$ B) $\frac{1}{3}$ C) $\frac{2}{3}$ D) $\frac{3}{8}$ ☒ E) $\frac{1}{6}$

$$\frac{21}{32} \times \frac{16}{63} = \frac{\cancel{21}^1}{\cancel{63}_3} \times \frac{\cancel{16}^1}{\cancel{32}_2} = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$$

Q28. $\left(\frac{18}{35}\right) \times \left(\frac{21}{24}\right) = ?$

- A) $\frac{3}{5}$ B) $\frac{3}{10}$ C) $\frac{7}{20}$ D) $\frac{9}{40}$ ☒ E) $\frac{9}{20}$

$$= \frac{18}{35} \times \frac{21}{24}$$

$$= \frac{9}{20}$$

Q29. $\left(\frac{9}{14}\right) \times \left(\frac{28}{45}\right) \times \left(\frac{15}{8}\right) = ?$

- A) $\frac{4}{3}$ ☒ B) $\frac{3}{4}$ C) $\frac{5}{6}$ D) $\frac{2}{3}$ E) $\frac{9}{8}$

$$\frac{9}{14} \times \frac{28}{45} \times \frac{15}{8} = \frac{9}{45} \times \frac{28}{14} \times \frac{15}{8}$$

$$= \frac{1}{5} \times 2 \times \frac{15}{8} = \frac{3}{4}$$

Q30. $\left(2\frac{1}{3}\right) \times \left(\frac{3}{7}\right) = ?$

- A) $\frac{3}{2}$ B) $\frac{7}{3}$ C) $\frac{3}{7}$ D) $\frac{2}{7}$ ☒ E) 1

$$= \left(2\frac{1}{3}\right) \times \left(\frac{3}{7}\right)$$

$$= \left[\frac{(2 \times 3) + 1}{3}\right] \times \frac{3}{7}$$

$$= \frac{7}{3} \times \frac{3}{7} = 1$$

Q31. $\left(\frac{5}{8}\right) \div \left(\frac{10}{21}\right) = ?$

- ✓ A) 21/16 B) 16/21 C) 5/16 D) 2/3 E) 3/2

$$= \frac{5}{8} \div \frac{10}{21}$$

$$= \frac{5}{8} \times \frac{21}{10} \quad (\text{Reciprocal})$$

$$= \frac{21}{16}$$

Q32. $\left(\frac{13}{20}\right) \div \left(\frac{26}{35}\right) = ?$

- A) 5/7 ✓ B) 7/8 C) 8/7 D) 13/14 E) 14/13

$$= \frac{13}{20} \div \frac{26}{35}$$

$$= \frac{13}{20} \times \frac{35}{26}$$

$$= \frac{7}{8}$$

either $\times$ to $\div$
or $\div$ to $\times$ hoga,
then value Reciprocal
ho jayegi

Q33. $\left(4\frac{1}{2}\right) \div \left(1\frac{1}{5}\right) = ?$

- ✓ A) 15/4 B) 4/15 C) 5/3 D) 3/5 E) 9/4

$$= \left(4\frac{1}{2}\right) \div \left(1\frac{1}{5}\right)$$

$$= \left(\frac{8+1}{2}\right) \div \left(\frac{5+1}{5}\right)$$

$$= \frac{9}{2} \div \frac{6}{5}$$

$$= \frac{9}{2} \times \frac{5}{6} = \frac{15}{4}$$

Q34. $\left(\frac{5}{6}\right) \div \left(\frac{2}{3}\right) \div \left(\frac{5}{4}\right) = ?$

A) $4/5$

B) 2

C) $1/2$ D) $5/4$

✓ E) 1

$$= \frac{5}{6} \div \frac{2}{3} \div \frac{5}{4}$$

$$= \frac{\cancel{5}}{6} \times \frac{3}{\cancel{2}} \times \frac{4}{\cancel{5}} = \frac{\cancel{5}}{\cancel{5}_2} \times \frac{\cancel{4}^2}{\cancel{2}} = \frac{2}{2}$$

$$= 1$$

Q35. $\left(\frac{2}{5}\right) \div \left(\frac{4}{15}\right) + \frac{1}{3} = ?$

A) $11/9$ B) $7/6$ C) $9/6$ ✓ D) $11/6$ E) $5/6$

$$\left(\frac{2}{5} \div \frac{4}{15}\right) + \frac{1}{3}$$

$$= \left(\frac{2}{5} \times \frac{15}{4}\right) + \frac{1}{3}$$

$$= \frac{3}{2} + \frac{1}{3} = \frac{9+2}{6} = \frac{11}{6}$$

{ BODMAS rule,
pehle Division,
then Addition, }

[LCM of 2 and 3
is 6]

Q36. $\left(\frac{7}{12}\right)$ of 132 = ?

A) 66

B) 72

✓ C) 77

D) 84

E) 88

$$= \frac{7}{12} \times 132$$

$$= \frac{7}{\cancel{12}} \times 11 \times \cancel{12} = 77$$

Q37. $\left(\frac{5}{8}\right)$ of 144 = ?

- A) 80 ☒ B) 90 C) 85 D) 95 E) 100

$$= \frac{5}{8} \times 144$$

$$= \frac{5}{\cancel{8}} \times (\cancel{8} \times 18) = 5 \times 18 = 90$$

Q38. $\left(\frac{3}{5}\right)$ of 250 - $\left(\frac{2}{3}\right)$ of 150 = ?

- A) 40 ☒ B) 50 C) 60 D) 70 E) 80

$$= \left(\frac{3}{5} \times 250\right) - \left(\frac{2}{3} \times 150\right)$$

$$= (3 \times 50) - (2 \times 50)$$

$$= 150 - 100 = 50$$

Miscellaneous

Q39. $\left\{\left(\frac{3}{4} + \frac{2}{3}\right)\right\} \div \left\{\left(\frac{5}{6} - \frac{1}{2}\right)\right\} = ?$

- ☒ A) 17/4 B) 4/17 C) 17/8 D) 7/4 E) 34/7

$$= \left\{\frac{3}{4} + \frac{2}{3}\right\} \div \left\{\frac{5}{6} - \frac{1}{2}\right\}$$

$$= \left\{\frac{9+8}{12}\right\} \div \left\{\frac{5-3}{6}\right\}$$

$$= \frac{17}{12} \div \frac{2}{6} = \frac{17}{\cancel{12}_2} \times \frac{\cancel{6}_2}{2} = \frac{17}{4}$$

Q40. $\left\{\left(\frac{7}{8} - \frac{3}{10}\right)\right\} \div \left\{\left(\frac{1}{4} + \frac{1}{5}\right)\right\} = ?$

A) 23/9 B) 18/23 C) 23/40 D) 23/18 E) 5/9

$$= \left\{\frac{7}{8} - \frac{3}{10}\right\} \div \left\{\frac{1}{4} + \frac{1}{5}\right\}$$

$$= \left\{\frac{70-24}{80}\right\} \div \left\{\frac{5+4}{20}\right\}$$

$$= \frac{46}{80} \div \frac{9}{20} = \frac{46}{80} \times \frac{20}{9} = \frac{23}{18}$$

Q41. $\left\{\left(\frac{5}{12}\right) \div \left(\frac{1}{6}\right)\right\} - \left\{\left(\frac{3}{4}\right) \div \left(\frac{9}{16}\right)\right\} = ?$

A) 11/6 B) 5/6 C) 1/6 D) 7/6 E) 2/3

$$= \left\{\frac{5}{12} \div \frac{1}{6}\right\} - \left\{\frac{3}{4} \div \frac{9}{16}\right\}$$

$$= \left(\frac{5}{12} \times \frac{6}{1}\right) - \left(\frac{3}{4} \times \frac{16}{9}\right)$$

$$= \frac{5}{2} - \frac{4}{3} = \frac{15-8}{6} = \frac{7}{6}$$

Q42. $\left\{1 \div \left(\frac{1}{2} + \frac{1}{3}\right)\right\} + \left\{1 \div \left(\frac{1}{3} + \frac{1}{6}\right)\right\} = ?$

A) 16/5 B) 8/5 C) 6/5 D) 18/5 E) 10/5

$$= \left\{1 \div \left(\frac{3+2}{6}\right)\right\} + \left\{1 \div \left(\frac{2+1}{6}\right)\right\}$$

$$= \left\{1 \div \frac{5}{6}\right\} + \left\{1 \div \frac{3}{6}\right\}$$

$$= \left(1 \times \frac{6}{5}\right) + \left(1 \times \frac{6}{3}\right)$$

$$= \frac{6}{5} + 2 = \frac{6+10}{5} = \frac{16}{5}$$

Q43. $\left\{\left(\frac{3}{5}\right) \div \left(\frac{1}{2}\right)\right\} \div \left\{\left(\frac{4}{7}\right) \div \left(\frac{2}{3}\right)\right\} = ?$

- A) $\frac{5}{6}$ B) $\frac{5}{7}$ C) $\frac{7}{6}$ D) $\frac{6}{7}$ ☒ E) $\frac{7}{5}$

$$= \left\{\frac{3}{5} \div \frac{1}{2}\right\} \div \left\{\frac{4}{7} \div \frac{2}{3}\right\}$$

$$= \left\{\frac{3}{5} \times \frac{2}{1}\right\} \div \left\{\frac{4}{7} \times \frac{3}{2}\right\}$$

$$= \left(\frac{6}{5}\right) \div \left(\frac{6}{7}\right) = \frac{6}{5} \times \frac{7}{6} = \frac{7}{5}$$

Q44. $\left[\left(\frac{3}{5}\right) \div \left\{1 - \left(\frac{2}{5}\right)\right\}\right] \times \left[\left(\frac{7}{9}\right) \div \left\{1 - \left(\frac{2}{3}\right)\right\}\right] = ?$

- ☒ A) $\frac{7}{3}$ B) $\frac{3}{7}$ C) $\frac{7}{9}$ D) $\frac{9}{7}$ E) 1

$$= \left[\frac{3}{5} \div \left(\frac{5-2}{5}\right)\right] \times \left[\frac{7}{9} \div \left(\frac{3-2}{3}\right)\right]$$

$$= \left[\frac{3}{5} \times \frac{5}{3}\right] \times \left[\frac{7}{9} \times \frac{3}{1}\right]$$

$$= 1 \times \frac{7}{3} = \frac{7}{3}$$

Q45: $3\frac{2}{5} + 4\frac{3}{10} + 2\frac{1}{2} = 6\frac{9}{10} + ? + 1\frac{11}{20}$

- A) $1\frac{1}{4}$ ☒ B) $1\frac{3}{4}$ C) $2\frac{1}{4}$ D) $2\frac{3}{4}$ E) None of these

$$3 + \frac{2}{5} + 4 + \frac{3}{10} + 2 + \frac{1}{2} = 6 + \frac{9}{10} + x + 1 + \frac{11}{20}$$

$$(3+4+2) + \frac{2}{5} + \frac{3}{10} + \frac{1}{2} = (6+1) + x + \frac{9}{10} + \frac{11}{20}$$

$$(9-7) + \frac{2}{5} + \frac{3}{10} + \frac{1}{2} - \frac{9}{10} - \frac{11}{20} = x$$

$$2 + \frac{8+6+10-18-11}{20} = x$$

$$2 + \left(-\frac{5}{20}\right) = x$$

$$2 - \frac{1}{4} = x$$

$$x = 1 + \frac{3}{4}$$

$$1 + \left(1 - \frac{1}{4}\right) = x$$

$$x = 1\frac{3}{4}$$

Q46. $9\frac{1}{6} - 7\frac{5}{8} + 6\frac{1}{3} = \frac{31}{8} + ?$

A) $7/6$

B) 6

C) $3/2$

✓ D) 4

E) None of these

$$9 + \frac{1}{6} - 7 - \frac{5}{8} + 6 + \frac{1}{3} = 3 + \frac{7}{8} + x$$

$$\left\{ \frac{31}{8} = 3\frac{7}{8} \right\}$$

$$9 + \frac{1}{6} - 7 - \frac{5}{8} + 6 + \frac{1}{3} - 3 - \frac{7}{8} = x$$

$$(9 - 7 + 6 - 3) + \frac{1}{3} + \left(-\frac{5}{8} - \frac{7}{8}\right) + \frac{1}{6} = x$$

$$5 + \left(\frac{1}{6} + \frac{1}{3}\right) + \left(-\frac{12}{8}\right) = x$$

$$5 + \frac{1}{2} - \frac{3}{2} = x$$

$$5 + \left(\frac{1-3}{2}\right) = x$$

$$x = 5 - 1 = 4$$

Q47. $9\frac{2}{9} + 5\frac{1}{18} - \frac{13}{18} = \frac{488}{?}$

A) 54

B) 18

C) 72

✓ D) 36

E) None of these

$$9 + \frac{2}{9} + 5 + \frac{1}{18} - \frac{13}{18} = \frac{488}{x}$$

$$(9 + 5) + \frac{2}{9} + \left(\frac{1}{18} - \frac{13}{18}\right) = \frac{488}{x}$$

$$14 + \frac{2}{9} - \frac{12}{18} = \frac{488}{x}$$

$$14 + \frac{4-12}{18} = \frac{488}{x}$$

$$14 - \frac{8}{18} = \frac{488}{x}$$

$$14 - \frac{4}{9} = \frac{488}{x}$$

$$\frac{126-4}{9} = \frac{488}{x}$$

$$\frac{122}{9} = \frac{488}{x}$$

$$x = \frac{488 \times 9}{122}$$

$$x = 36$$

Q48: $? \times 4\frac{3}{5} = 6\frac{1}{4} \times 3\frac{5}{6} \times 2\frac{2}{3}$

A) $7\frac{1}{5}$

B) $8\frac{1}{3}$

C) $9\frac{2}{5}$

D) $13\frac{8}{9}$

E) None of these

$$x \times \frac{23}{5} = \frac{25}{4} \times \frac{23}{6} \times \frac{8}{3}$$

$$x = \frac{25}{4} \times \frac{\cancel{23}}{6} \times \frac{8}{3} \times \frac{5}{\cancel{23}}$$

$$x = \frac{125}{9}$$

$$x = 13\frac{8}{9}$$

Q49: $5\frac{1}{2} \div 11\frac{2}{5} \div 2\frac{2}{19} = ?$

A) $7/12$

B) $11/48$

C) $13/24$

D) $21/20$

E) None of these

$$\left(5\frac{1}{2}\right) \div \left(11\frac{2}{5}\right) \div \left(2\frac{2}{19}\right)$$

$$\frac{11}{2} \div \frac{57}{5} \div \frac{40}{19}$$

$$\frac{11}{2} \times \frac{5}{\cancel{57}} \times \frac{\cancel{19}}{40}$$

{ Reciprocal }

$$= \frac{11}{48}$$

Q50. $1 - \frac{1}{1 + \frac{1}{2 + \frac{1}{3 + \frac{1}{4 + \frac{1}{2}}}}} = ?$

(A) 29/96

(B) 9/32

(C) 31/96

(D) 25/96

(E) 29/94

$$= 1 - \frac{1}{1 + \frac{1}{2 + \frac{1}{3 + \frac{1}{4 + \frac{1}{2}}}}}$$

$\left\{ 4 + \frac{1}{2} = \frac{9}{2} \right\}$

$$= 1 - \frac{1}{1 + \frac{1}{2 + \frac{1}{3 + \frac{2}{9}}}}$$

$\left\{ 3 + \frac{2}{9} = \frac{29}{9} \right\}$

$$= 1 - \frac{1}{1 + \frac{1}{2 + \frac{9}{29}}}$$

$$= 1 - \frac{1}{1 + \frac{67}{29}}$$

$$= 1 - \frac{1}{1 + \frac{29}{67}}$$

$$= 1 - \frac{1}{96/67}$$

$$= 1 - \frac{67}{96} = \frac{29}{96}$$